Improving Transient Stability of an Islanded Microgrid Using PV Based Virtual Synchronous Machines

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***Abstract*—Renewable energy integration is rapidly increased in this decade to achieve clean energy production targets. Inverters are commonly used to integrate renewable energy sources, such as wind and solar to the power system. Conventional power generation such as Coal, Nuclear, Hydro are based on synchronous generators. Due to the large integration of renewable power plants, the power system is moving towards an inverter dominant power system. This can make a significant impact on the system inertia response. Inertia is given from the rotating mass of the conventional generators but, inverters cannot contribute to system inertia as there are no rotating masses. Lack of inertia will weaken the stability of the system under disturbances. As a result, system frequency will fluctuate beyond its limits. This limits the integration of renewable sources to grids. However, inverters can emulate inertia response because of the fast response and the high controllability. This paper presents two novel controllers that can be used to emulate the inertia response. One controller is for the virtual inertia controlling and the other controller is for energy management. Results are obtained and analyzed in a simulated PV-Hydro microgrid under supply-demand mismatch.**

Keywords—Energy Management, Renewable integration, Transient frequency stability, Virtual Inertia

I. INTRODUCTION

The world is moving more towards renewable energy resources because of its eco-friendly behavior. There are many agreements, such as the Paris agreement in December 2015 which encourages renewable energy integration [1]. According to these agreements many countries have agreed to increase electricity generation through renewable sources to improve clean energy production. Countries like Denmark, Ireland and Germany are rapidly installing renewable power plants. They are operating with an annual penetration of more than 20% at the national level [2]. United Kingdom has set a target of 15% of their electricity generation being renewable by 2020 [3]. Sri Lanka has also set a target of 100% renewable power generation by the year of 2050 [4].

Most of these renewable sources are found as distributed power sources. Therefore, the concept of micro grids becomes more familiar. A micro grid is a group of local grids with a number of renewable energy sources, energy storage, and domestic loads, which can be identified as an independent system with the capability to operate in either grid connected or isolated mode, thus reducing a huge burden on the utility grids [5] ,[6].

Most of the dominant renewable power sources (solar and wind) deliver power via inverters. Therefore, inverter-based power sources are going to be a dominant part in future grids [7]. Because of the large penetration of renewable sources, inverter dominant power generation has been rising rapidly. Many rural areas use small hydro power plants as their main source of power generation. More than 45,000 small hydro plants have been installed to supply energy for more than million rural population in China [8]. However, the biggest issue with small hydro plants is that it's inability to produced power to match the demand is in dry seasons. To overcome this, most the small hydro based microgrids are integrated with solar PVs to guarantee a reliable power supply throughout of the year. Furthermore, Hydro-PV systems are getting more economically feasible as solar PVs are getting more efficient and less expensive yearly.

Even though inverters help to connect more renewable sources to the system, since it does not supply any mechanical inertia, it compromises the frequency stability of the system. Installing more inverter dominant power plants will further reduce the inertial response of the whole system, which will make the system more vulnerable for frequency fluctuations during large disturbances. Consequently, the system operators limits the inverter integrated renewable generation to the system.

Frequency is an indicator of balance between generation and demand of active power. Primary frequency control is used to control the frequency by adjusting the prime movers with a droop control mechanism. The primary frequency response is the turbine governor action or governor like action. Depending on the input power variation to the governor, the generator will increase or decrease the active power generation proportional to system frequency deviation. However, it takes 10s to 30s to initiate the primary response. Until the primary response inertia response controls the frequency response. Inertia helps to keep the frequency within standard limits until the primary response is involved [9].

Even though inverters do not have inertia, inverters can emulate the inertia response due to their high controllability and fast response [10]. There are many virtual inertia topologies available in literature. Basic topology of emulating inertia though inverters is shown in Fig 1. The DC Source represents the renewable energy source such as solar PVs. The DC-DC converter is used to produce a constant voltage at the DC link. Then, using virtual inertia topologies, gate signals are produced to emulate the behavior of the inertia response. Inertia emulating techniques are discussed in the following subsections.

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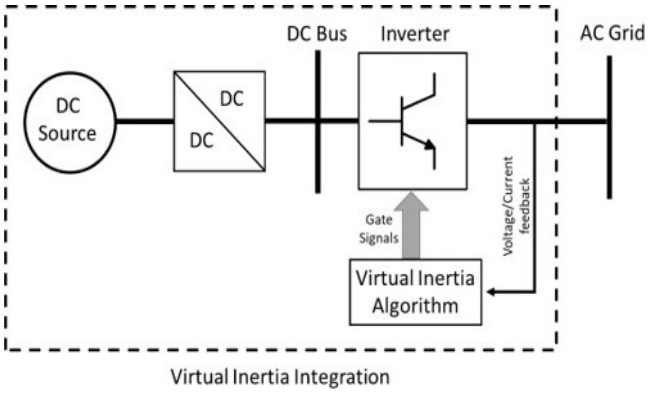


Fig. 1.Virtual inertia integration mechanism

A. Synchronverters

Synchronverter topology emulates the same ideal behavior of a synchronous generator from the point of view of the grid with the aim of integrating renewable energy sources without having a major change in the traditional power system [11],[12],[13]. Traditional grid controls can be followed while integrating more inverter dominant power generation [14]. Since this synchronverter models the exact behavior of a synchronous generator the model is very complex. Therefore higher computational power is required, and due to the higher complexity, response time may increase.

B. Swing equation based

Swing equation based topology rather than modeling the exact behavior of a synchronous generator, the swing equation (1) is modeled and solved in each control cycle to emulate the inertia response. This has a reduced complexity compared to the Synchronverter model. Therefore, the high computational power requirement is eliminated [15],[16]. However, parameters must be carefully selected to avoid unwanted frequency oscillations.

$$P_{in} - P_{out} = J\omega_m \frac{\mathrm{d}\Delta\omega}{\mathrm{dt}} + D_p\Delta\omega \tag{1}$$

$$\Delta\omega = \omega_m - \omega_0 \tag{2}$$

Where J and D_p represent the moment of inertia and damping constant properties respectively, P_{in} is the transferred power to the inverter, P_{out} is the output power from the inverter, ω_m is virtual angular frequency and ω_0 is reference angular frequency. Phase Lock Loop (PLL) is used to measure the frequency and from that $\Delta\omega$ is calculated. In the swing equation-based topology J and D_p are always kept as constant. However, J and D_p can be dynamically changed since they are virtually emulated properties.

C. Frequency-power response-based topology

Frequency-power response-based topology is the simplest approach of emulating the inertia. This topology is not intercorporated with any modelling of synchronous generator. This topology provides dynamic frequency regulation based on the deviation of the measured frequency and the deviation of the rate of change of the frequency.

$$P_{in} - P_{out} = K_1 \frac{\mathrm{d}\Delta\omega}{\mathrm{dt}} + K_p\Delta\omega \tag{3}$$

$$\Delta\omega = \omega_m - \omega_0 \tag{4}$$

Where K_1 and K_p are represent the inertia and damping properties respectively. Damping constant helps to regain frequency to its nominal value and reduces the frequency nadir. Inertia constant slower the Rate of change of frequency (ROCOF) by providing dynamic frequency response based on system frequency deviations.

D. Droop-based approaches

Droop based approaches [17],[18] does not involve modelling the behavior of synchronous generators. Equation (5) is used to model the droop based approaches.

$$\omega_m = \omega_0 - \omega_p(P_{out} - P_{in}) \tag{5}$$

Where, ω_m is virtual angular frequency and ω_0 is reference angular frequency, P_{out} is the measured active power output from the inverter P_{in} is the reference active power and ω_p is the active power droop. Disadvantages of this method are slow transient response, poor dynamic inertial response support and deviation from the synchronous generator model.

Since the inertia is virtually implemented there is an added advantage. Inertia behavior is defined by the rotating masses and inertia is fixed for a generator. However, when modeling the virtual inertia, it is not required need to be a constant. In virtual inertia, inertia constant can be dynamically changed and the inertia response can be further improved.

Disturbance in the system normally interrupts the Maximum power point tracking (MPPT) operation of solar PVs. This reduced the efficiency of the solar PV under interruptions.

II. Proposing Controllers

This paper presents two novel controllers. Unlike aforementioned methods, which are based on fixed virtual inertia and where MPPT operation is interrupted while emulating the inertia, this paper presents a novel method to emulate variable inertia while another novel controller is introduced to support the continuous operation of MPPT.

Schematic of the system with proposed architecture is shown in Fig 2 and it involves designing of two novel controllers:

1. Virtual Inertia controller
2. Energy Management controller

A. Virtual Inertia Controller

Virtual inertia controller is based on the frequency-power response based. The controller is based on the equations (6) and (7).

$$P_{VSM} = K_1 \frac{\mathrm{d}\Delta\omega}{\mathrm{dt}} + K_D\Delta\omega \tag{6}$$

$$\Delta\omega = \omega_m - \omega_{Ref} \tag{7}$$

where P_{VSM} emulate the inertia response of a synchronous generator,$\omega_m$is virtual angular frequency, ω_{Ref} ($2\pi60$ rad/s or $2\pi50$ rad/s) is reference angular frequency , K_1 represent the inertia properties of the virtual synchronous generator and K_D represent the damping properties of the synchronous generator.

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Power will generate or absorb according to the frequency deviation and the ROCOF. Clearly K_1 works as an amplifier for ROCOF. Therefore, P_{VSM} should be able to exchange its maximum power when the maximum allowable ROCOF occurs. Also, K_1 should be a negative constant in order to counteract with the direction of ROCOF. K_D represents the damping constant. It acts as a barrier to the frequency change. K_D which is a negative constant helps to return the frequency to its nominal value. According to the equation (7) when there is no angular frequency deviation ($\Delta\omega$= 0) P_{VSM} is also zero. So, the VSM controller does not interrupt during the nominal operation. Therefore, this controller can be lumped up with the normal inverter controller it will automatically trigger when there is a frequency deviation detected and again it will automatically switch off when the frequency return to the rated frequency. In literature K_1 and K_D are calculated using equation (8) and (9) [19].

$$K_1 = \frac{P_{VSG_NOM}}{ROCOF_{MAX}} \tag{8}$$

$$K_1 = \frac{P_{VSG_NOM}}{\Delta\omega_{MAX}} \tag{9}$$

where P_{VSG_NOM} is the nominal power rating of the VSM. Literature suggest that to select P_{VSG_NOM} as 5% of the total capacity of the microgrid in-order give more effective results [20].

1) Stability criteria with load angle curve: Fig. 3 shows load angle curve of a typical generator for a connection of a large load. Point A is the original operating point at P_1 power. Then suddenly due to the connection of load, operating point jumps to point B, where power is P_2.

When the operating point moves along the curve from point A to B, Δω is positive and the machine is in acceleration. The target is to fix the new operating point at B as soon as possible. when the load is connected, the operating point does not move to the point B right away. Instead, it will go until point C in the first cycle. The time taken to move along the curve from A to B can be reduced by introducing high inertia. High inertia means high kinetic energy. Therefore, time from A to B can be made much shorter by releasing the required amount of kinetic energy. From B to C along the curve, Δω is still positive but the machine is on deceleration mode. Having higher inertia will not help in this situation. Higher inertia will take point C further away from point B. For this situation, low inertia will effectively reduce the length between B and C in the curve (point C gets much closer to point B). From C to B along the curve, Δω is negative but the machine is in acceleration mode. Thus, inertia has to be increased to reduce the time from C to B. After passing point B, again Δω is negative and the machine is in a deceleration mode and having a low inertia will help in this part. Low inertia will reduce the amount it deviates from the point B again. Thus, having a varying inertia supports the system to be transiently stable much faster.

However, implementation of the above mechanism in conventional generators is not feasible, because connecting and disconnecting masses is not efficient and practical. However, it is possible with the concept of virtual inertia. Each and every operating region needs to be identified precisely in order to implement the above controller. The Identification of regions can be easily done by using Δω and ROCOF. TABLE I summarizes the identification method of the areas in the power angle curve using the behavior of Δω and ROCOF. After identifying each section, inertia constant (KI) can be assigned to the corresponding section.

2) Variable inertia constant controller: System frequency is measured using a PLL. As shown in Fig. 4, frequency deviation is calculated by comparing the measured frequency with the reference frequency, which is then multiplied by 2π in order to calculate the angular frequency deviation. Then the ROCOF is calculated by finding the derivative of the change in angular frequency.

3) Virtual inertia controller implementation: Equation (6) is modelled in the controller as shown in Fig. 5. Inertia constant is fed from the variable inertia constant controller. Deviation of angular frequency is multiplied from the damping constant and added to the multiplication of the variable inertia constant and the derivative of the deviation of angular frequency. This generates the required active power that needs to be exchanged from the inverter to enhance the inertia support. P_{VSM} is the active power reference of the inverter.

4) Current controller: Current controller produces a current reference based on the active power reference produced by the virtual inertia controller. Direct-quadrature(d-q) based rotating frame linear current controlling is used [21]. Based on the power reference, the required current is calculated by using (10).

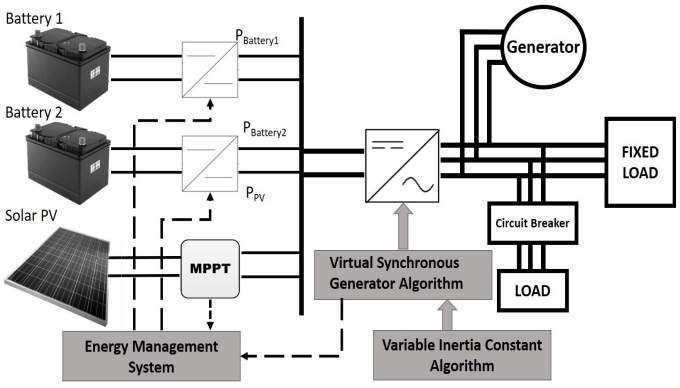


Fig. 2. Proposed architecture

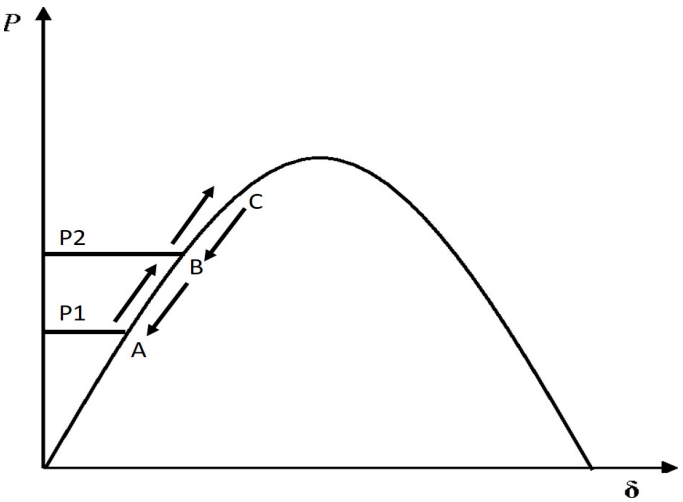


Fig. 3. Load angle cure for a sudden demand increase situation

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TABLE I.
SEGMENTS IDENTIFICATION OF THE LOAD ANGLE CURVE

Segment	Δω	d(Δω)/dt	Mode	KI
A to B	> 0	> 0	Acceleration	High
B to C	> 0	< 0	Deceleration	Low
C to B	< 0	> 0	Acceleration	High
B to A	< 0	< 0	Deceleration	Low

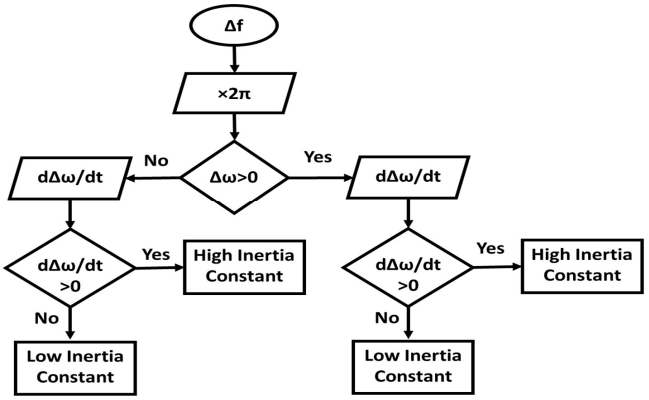


Fig. 4. Variable inertia constant controller

$$I_d = \frac{2}{3}\left(\frac{V_d P_{VSM} - V_q Q}{V_d^2 - v_q^2}\right) \quad (10)$$

$$I_q = 0 \quad (11)$$

Where I_d and I_q are respectively d and q axis required current components of the inverter respectively, V_d and V_q are d and q component of the grid voltage respectively and Q is the reactive power reference. Reactive power reference is set to zero as active power is the only interest for the inertia response. Then, in the secondary control loop from the equation (12) is used to calculate the d-q axis voltage references.

$$\begin{bmatrix} V_{d,New} \\ V_{q,New} \end{bmatrix} = L\frac{d}{dt}\begin{bmatrix} I_d \\ I_q \end{bmatrix} + R\begin{bmatrix} I_d \\ I_q \end{bmatrix} + \omega L\begin{bmatrix} -I_q \\ I_d \end{bmatrix} + R\begin{bmatrix} e_d \\ e_q \end{bmatrix} \quad (12)$$

Using the synchronization angle θ, d-q components are coupled back to produce the three phase reference voltages, which are compared against a high frequency triangular waveform to produce Pulse Width Modulated (PWM) signals to MOSFETs in the inverter.

B. Energy Management System (EMS)

As shown in the Fig 2, EMS contains two sets of batteries and separate controllers for each battery. The purpose of this design is to enhance the response time. Suppose there is a situation that needs to change the operation status from charging to discharging state immediately. In such a situation, if only one battery is used, the state of battery operation needs to be changed immediately from charging mode to discharging mode. This sudden change of states will introduce unwanted transients and is harmful for the battery lifetime. By introducing two batteries such situations can be avoided, because while one battery is on the charging mode the other battery can be set to discharging configuration. When the change of states is required, charging battery can be switched off and the discharging battery can be switched on. This will eliminate unwanted transients, and increase the battery life time.

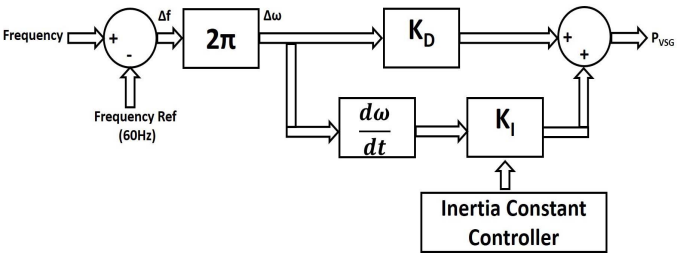


Fig. 5. Virtual inertia controller

EMS takes inputs: P_{VSM} generated from the VSM controller and power measurement from the PV system, which produces two control signals for each battery controller.

As shown in Fig. 6, EMS first checks the frequency from PLL and if the grid frequency is higher than 60 Hz, both batteries will go to charging mode and the power that needs to be absorbed is given from the (13).

$$P_{Battery_1} = P_{Battery_2} = +\left[\frac{P_{VSM}+P_{pv}}{2}\right] \quad (13)$$

If the power requirement is higher than the power generated by the solar PV and the required additional power is less than the power of a single battery, one battery will discharge, and the other battery will be turned off. Power reference for the battery is given by (14).

$$P_{Battery_1} = -\left[P_{VSM} - P_{pv}\right] \quad (14)$$

If the power requirement is higher than the power generated by the solar PV and the required additional power is higher than the power of a single battery, both batteries will be discharged. Require power for each battery is given by (15).

$$P_{Battery_1} = P_{Battery_2} = -\left[\frac{P_{VSM}-P_{pv}}{2}\right] \quad (15)$$

If the power requirement is less than the power generated by the solar PV and the required additional power is less than the power of a single battery, one battery will be discharged, and the other battery will be turned off. Power reference for the battery is given by (16).

$$P_{Battery_1} = +\left[P_{pv} - P_{VSM}\right] \quad (16)$$

If the power requirement is less than the power generated by the solar PV and the required additional power is higher than the power of a single battery, both batteries will be discharged. Required power for each battery is given by (17).

$$P_{Battery_1} = P_{Battery_2} = +\left[\frac{P_{pv}-P_{VSM}}{2}\right] \quad (17)$$

III. RESULTS

Two cases were simulated.

1. Case 1: Load adding
2. Case 2: Load disconnecting

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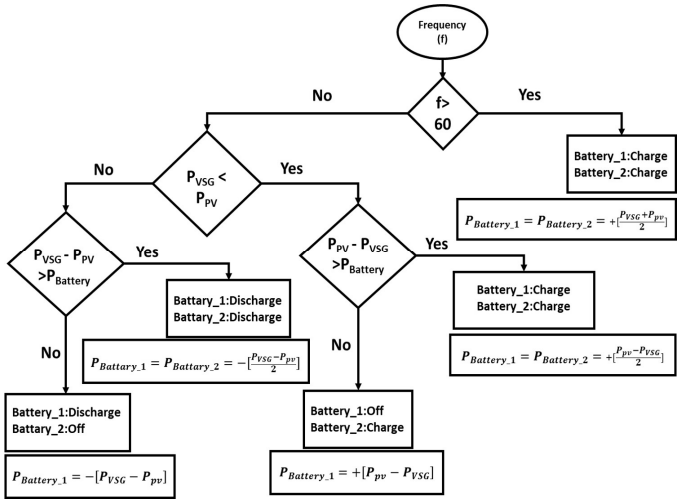


Fig. 6. Energy Management System Controller

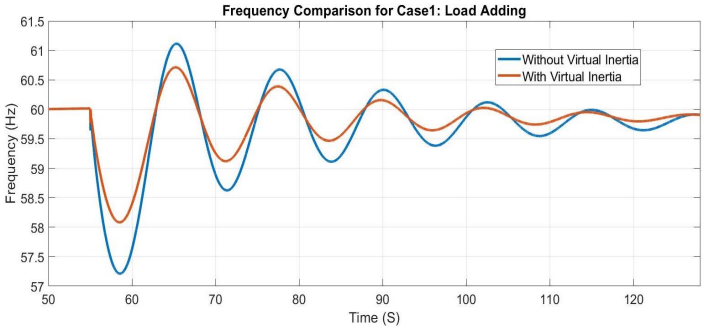


Fig. 7. Frequency comparison with and without VSM and EMS for Case 1

A. Case 1: Load Adding

This case represents a scenario where load is being connected. Initially, breaker is in open state so that the total demand is equal to the fixed load (23 MW). Once the system has reached its steady state, second load (3 MW) is connected by closing the breaker. Transient response is observed with and without the proposed VSM controller and the EMS. Frequency has dropped to 57.21 Hz in the first cycle without the VSM and EMS controllers as shown in Fig 7. However, with VSM and EMS controllers frequency has reduced only down to 58.08 Hz.

B. Case 2: Load Disconnecting

This case represents a sudden load discharging scenario. Initially breaker is in close position with both loads being connected. Once the system has reached its steady state, breaker is opened and the 3 MW load will disconnected suddenly. Transient response is observed with and without the proposed VSM and EMS controllers.

As shown in Fig. 8 frequency is increased up to 63.01 Hz immediately following the disconnection in the case where conventional controls are used. However, with the proposed VSM and EMS as shown in Fig 8, increment of frequency has reduced to 62.02 Hz.

It is evident from Fig.9 that the PV can operate at its maximum power point and continually produce power without any interruption, even during frequency fluctuations due to the action of the newly proposed controlling algorithm.

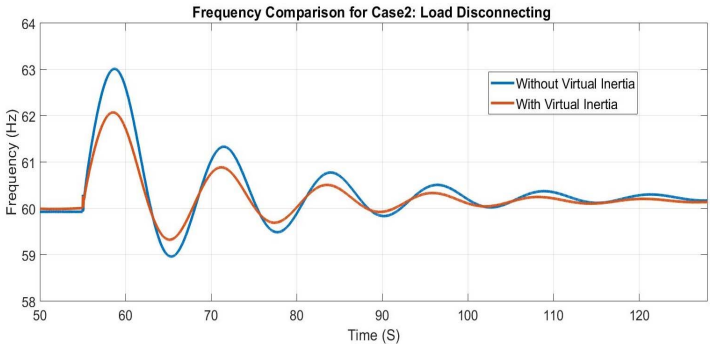


Fig. 8. Frequency comparison with and without VSM and EMS for Case 2

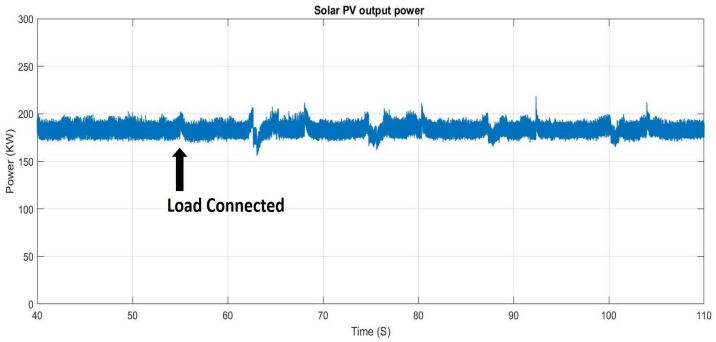


Fig. 9. Output power of the PV system under disturbance

IV. Evaluation of Controllers

VSM and EMS algorithms has enhanced the transient response of the system under sudden disturbances by improving the frequency recovery. This means that the system inertia response has been improved by the VSM and EMS controllers. Therefore, with VSM and EMS controllers, stress on the primary control has been reduced. Thus, system recover faster compared to the system with conventional controls.

Energy management controller makes it possible to run the solar power plant at its maximum power point in most of the time, even during disturbances.

Lack of inertia in conventional inverter driven power generation sources imposes limits on the capacity of converter-based renewable power generation that could be connected to a system. However, with the proposed VSM and EMS controller, inverter driven power plants can emulate the inertia response. Having a variable inertia constant is an added advantage, which is not present even in traditional generators.

Implementation of the proposed controller does not require much changes for the existing systems. The proposed control algorithm can be easily adopted to the PV inverters which, are integrated with energy storages.

V. Conclusion

Future power gird inclines more towards eco-friendly renewable power generation. Most of these renewable power generation are inverter dominant. Lack of inertia in inverter dominant power generation degrades the system frequency stability under supply demand mismatches. Therefore, integrating inverter type renewable energy sources are limited by system operators. This paper proposed two novel control algorithms for inverter dominant power plants to emulate the inertia response successfully and to improve the continuous

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operation of MPPT during interruptions. Performance evaluation reveals that, with the variable inertia controller, inverters can dynamically enhance the inertia response. With this novel concepts, microgrids are expected to accommodate more inverter type renewable plants with reduced limitations. It will be a great support for the future of eco-friendly power generation.

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